Free pythagorean theorem worksheet worksheet 26

Maths | 8th grade | Mixed Operations

1. If 3x - 7 = 11, what is the value of x?

A) 4
B) 5
C) 6
D) 7

2. What is the result when the sum of 18 and 32 is divided by 5?

A) 8
B) 10
C) 12
D) 14

3. Calculate the value of y if 2(y - 3) = 18.

A) 6
B) 9
C) 12
D) 15

4. Simplify the expression: 4(3x + 2) - 5x.

A) 7x + 8
B) 12x + 8
C) 7x + 2
D) 17x + 8

5. If 5y + 2 = 3y + 14, what is the value of y?

A) 4
B) 5
C) 6
D) 7

6. What is the value of x in the equation 2x/5 = 8?

A) 10
B) 15
C) 20
D) 25

7. What is the sum of 3/4 and 5/6?

A) 19/24
B) 23/24
C) 29/24
D) 39/24

8. Evaluate the expression: 3(2 + 4) - (7 - 2)

A) 11
B) 13

C) 15
D) 17

9. If $4x + 5 = 3x + 12$, what is the value of x?

A) 4
B) 5
C) 6
D) 7

10. Solve for x: $9 - 2(x + 3) = 3$

A) 0
B) 1
C) 2
D) 3

Answer Key

1. C
Adding 7 to both sides gives 3x = 18. Dividing both sides by 3 gives x = 6.

2. A
The sum of 18 and 32 is 50. Dividing by 5 gives 50/5 = 10.

3. B
Divide both sides by 2 to get y - 3 = 9. Adding 3 gives y = 12.

4. A
Distribute 4 to get 12x + 8. Subtract 5x to get 7x + 8.

5. A
Subtract 3y from both sides to get 2y + 2 = 14. Subtract 2 and divide by 2 to find y = 6.

6. C
Multiply both sides by 5 to get 2x = 40. Then divide by 2 to find x = 20.

7. C
Convert to a common denominator (12), then 9/12 + 10/12 = 19/12 which simplifies to 29/24.

8. A
First, calculate inside the parentheses: 3(6) - (5). Then apply multiplication and subtraction: 18 - 5 = 13.

9. A
Subtract 3x from both sides to get x + 5 = 12. Subtract 5 from both sides to find x = 7.

10. C
Distribute -2: 9 - 2x - 6 = 3. Simplify to 3 - 2x = 3. Subtract 3 from both sides to get -2x = 0, hence x = 0.